Unit 7: Data Link Control and Protocol

1.Explain ARQ and stop & wait ARQ error control mechanism. (4+4) 2011 [S]

2.What is ARQ? How does stop and wait ARQ and sliding window ARQ deals with error? [5] 2011[F], [7] 2014 [S] ,2025 (New)

3.How ARQ differ from stop and wait ARQ? Explain in regard to their working mechanism with necessary figures. [7] 2012 [S]

4.What do you mean by data compression technique? Explain the different error detecting techniques. [8]

5.Explain how error can be controlled using stop and wait methods and sliding window method. Also mention the advantages and disadvantages for both the methods. [8] 2012 [F] ,2019 [F]

6.Explain the importance of error control and flow control in communication system. Briefly explain sliding window flow control technique. [8] 2013 [S] ,2018 [F],2023 [F] (old) ,2023 [F] (New)

7.State various data compression technique used in data communication. Explain one of them in detail. [7] 2013 [S], [5] 2013 [F]

8.Describe flow control and error control. How does Go-Back-N ARQ work? [8] 2018 [F],[7] 2025 [S] (New)

9. Why is stop-and-wait ARQ inefficient? Explain the major difference between Go-Back-N ARQ and selective-reject ARQ. [7] 2014 [F]

10. Why is flow control necessary? Explain sliding window flow control mechanism. [7] 2015 [F], 2022 [F]

11. Explain the advantages and disadvantages of stop-and-wait ARQ. [4] 2016 [F]

12. Why is flow control required in data Communication? Explain how lost and duplicate packets are handled by selective repeat request ARQ. [7] 2017 [F], 2023 (old) [F]

13. Explain the different error control mechanism using stop and wait methods. Compare and contrast Go-Back-N ARQ with selective repeat ARQ. [7] 2018 [S]

14. Define flow control and error control. Explain how stop-and –wait ARQ protocol operate. [7] 2019 [S]

15.Describe the goals of multiplexing. Differentiate between FDM and TDM. Assume that a voice channel occupies a bandwidth of 4 KHz. We need to multiplex to voice channels with guard bands of 500 Hz using FDM. Determine the bandwidth of the link. Also show the multiplexing scheme diagrammatically. [7] 2019 [S]

16. Discuss about Go-Back-N ARQ and selective repeat ARQ in brief. [5] 2020 [S]

17. What is ARQ? What are the two types of sliding window ARQ error control? Explain. [8] 2021 [F],2024 [F] (New)

18.Define flow control as a layered protocol model. Explain the mechanism of a flow control for an ARQ model. [8] 2023 [S]

19. What are data link layer design issues? How does Go-Back-N Automatic Repeat Request (ARQ) improve the stop-and-wait protocol? [7] 2024 [S]

20. When is the retransmission necessary? Explain about stop and wait ARQ protocol for the damaged frame. [7] 2024 [S] (New)

21. Write short notes on

- HDLC protocol [5] 2024 [S]
- HDLC Protocol Vs Point-to-Point protocol [5] 2024 [F]

22.Explain how sender-receiver synchronization issues are addressed in the sliding window protocol, focusing on Selective Repeat mechanisms. Discuss their advantages, disadvantages, and how they ensure reliable data transmission in high-speed networks. [8] 2024 [F]

23.When is retransmission necessary? Describe the two types of sliding window ARQ protocols and explain how they ensure reliable data transmission in noisy communication channels. [7] 2025 [S]

Unit 8: Multiplexing & Switching

1. Define switching. How is a signal forwarded in packet and message switching? What are the different techniques of multiplexing? [2+5] 2011 [S]

2.What are the different techniques of multiplexing? [2] 2011 [S]

3. Write short notes on

- Telephone services analog & digital. [5] 2011 [S]
- Circuit, packet and message switching [5] 2023 [F] (old)
- Private branch exchange (PBX) [5] 2015 [S]
- Analog hierarchy [5] 2015 [S] ,2024 [F]
- Packet switching and message switching [5] 2021 [S]
- Types of switching [5] 2023 [S]
- Digital hierarchy in telephone system [5] 2025 [S]

4. Describe the analog hierarchy in which groups of signals are successively multiplexed onto higher bandwidth lines. [5] 2011 [F]

5. Describe the function of synchronous TDM? What are its advantages over asynchronous TDM? [5] 2011 [F]

6. Compare the performance of circuit switching and packet switching. [5] 2011 [F]

7. Explain the term message switching, packet switching and circuit switching. [7] 2012 [S], 2012 [F],2019 [S], 2020 [F], 2021 [F]

8. Explain the term multiplexing and switching. What are the components of private branch exchange? [8] 2013 [S]

9. Define multiplexing. What are its applications? [5] 2013 [F]

10. Differentiate between FDM and TDM. Explain its advantages and disadvantages of packet switching. [8] 2014 [F]

11. Explain the need for multiplexing and switching in data communication. [7] 2014 [F]

12.What is packet switching? How can the input circuit can be multiplexed in analogy hierarchy? [7] 2014 [S]

13.Differentiate between circuit, packet and message switching. [8] 2015 [S], 2016 [F]

14. What is the significance of switching? Explain different multiplexing techniques briefly with application of each. [8] 2015 [F]

15. Explain wave-division multiplexing. [4] 2016 [F]

16. Explain TDM. Compare Virtual-circuit network with datagram networks. [7] 2016 [S], 2017 [F]

17. Differentiate between TDM and FDM. [5] 2017 [F]

18. What is multiplexing? Explain FDM hierarchy in telephone systems. [7] 2018 [F], 2023 [F] (old)

19. What are the different techniques of multiplexing? [7] 2018 [F]

20. Explain the multiplexing technique applied in digital telephony with their significant application and multiplexing hierarchy. [7] 2018 [S], 2020 [F], 2021 [F]

21. Write notes on analog services and its hierarchy? Compare and contrast different types of switching mechanism. [8] 2019 [F]

22. Why do we need multiplexing? Explain different multiplexing techniques with its relevant figures. Also explain about E1 telephony in brief. [10] 2020 [S]

23. Compare and contrast between circuit switching, packet switching and message switching. [8] 2022 [F], 2023 [F] (New)

24. Compare and contrast between virtual circuit networks and datagram network. [7] 2023 [S]

25. Explain the multiplexing technique applied in digital telephony with their significant application and multiplexing hierarchy. [7] 2023 [F] (New), 2024 [S]

26. Is switching and routing same? [4] 2023 [F] (New)

27.Briefly explain circuit switching and packet switching. [8] 2024 [S]

28.Explain the significance of switching? Also, Explain FDM hierarchy in telephone system. [7] 2024 [F]

29. What do you mean by multiplexing? How is TDM different from FDM? [7] 2024 [F]

30.Compare and contrast circuit switching, packet switching, virtual circuit packet switching. [8] 2025 [S]

31.What do you understand by PBX? Compare among circuit switching, message switching and packet switching. [8] 2025 [S] (New)

Unit 9: Data Encoding & Modulation

1. Explain the block diagram of PCM system. [6] 2012 [S]

2. Write short notes on

- QPSK modem or QPSK [5] 2011 [S], 2011 [F], 2012 [S], 2014 [F], 2015 [F], 2017 [F], 2018 [S],2024 [F]
- Encoding [5] 2011 [S]
- PCM [5] 2012 [S] , 2022[F]
- Delta modulation [5] 2012 [F]
- DQPSK [5] 2013 [S]
- Unipolar and bipolar encoding [5] 2013 [S]
- Standard of modem [5] 2013 [F]
- QAM modulation. [5] 2021 [F]
- Modem [5] 2022 [F]

3. What is the difference between encoding and modulation? (2013) What is digital-to-digital encoding? Where is it used? [7] 2011 [F], 2013 [S], 2013 [F]

4. Compare and contrast amplitude and frequency modulation techniques. [8] 2011 [F], 2012 [S]

5. Why do we use line coding? Line code the bit stream 8045h using HDB3, B8ZS, NRZ, RZ. [8] 2012 [F]

6. What is multilevel modulation? When is it necessary? Explain working concept of a QPSK modem. [7] 2012 [F], 2019 [F]

7. What is the necessity of modulation? Draw and explain block diagrams of PCM systems. [7] 2013 [S]

8. Encode the bit stream: 1100001011 using [7] 2013 [S]

i. Polar RZ,
ii. Polar NRZ,
iii. AMI
iv. Manchester
v. HDB3

9. What is biphase encoding? How does biphase overcome the limitation of NRZ codes? Explain with examples. [7] 2013 [F]

10. Differentiate between pulse code modulation and delta modulation. [4] 2014 [F], 2019 [S]

11. Differentiate between encoding and modulation. The sequence of the bit is 010111100. What will be the output if the following encoding schemes are used?

i) AMI

ii) NRZ

iii) NRZ-L

iv) RZ

v) Differential Manchester [7] 2014 [F]

12. Define modulation. Explain the use of modulation. [5] 2014 [S]

13. What is DC wonder? Using HDB3 technique generates the signal response for the code 11000011100001. [5] 2014 [S]

14. Define modem and explain the high-speed modem (56 modem). [5] 2014 [S],[7] 2015 [S]

15. Explain how HDB3 and B8ZS methods handle long runs of zeros which AMI fails to handle. Explain phase modulation. [8] 2015 [S]

16. How do we represent ASK and PSK modulation? Represent 1010101 using Manchester and AMI codes. [7] 2015 [S]

17. With detailed figures explain QPSK techniques? [8] 2015 [S]

18. How does BPSK differ from DPSK? How does Delta modulation overcome the demerit of PCM? Explain.

19. Differentiate between encoding and modulation. Encode the bit stream 1110000101 using

i. Polar RZ
ii. Polar NRZ
iii. AMI
iv. HDB3 [7] 2015 [F]

20. What do you mean by modulation and demodulation? Explain FM. [8] 2016 [F]

21. Explain different methods to convert analog signal into digital data. [7] 2016 [F]

22.Explain delta modulation. Enumerate QPSK modem in details. [8] 2016 [F]

23. Differentiate encoding and modulation. The sequence of the bit is 010111100. What will be the output of the following encoding schemes?

i) AMI

ii) NRZ

iii) NRZ-L

iv) RZ

v) Differential Manchester [8] 2016 [S]

24. What is a modem? Describe multilevel modulations. [7] 2016 [S]

25. Explain phase modulation and frequency modulation. [7] 2017 [F]

26. Differentiate between encoding and modulation with diagram. Explain QPSK technique. [8] 2018 [F], 2020 [F], 2023 [F] (old)

27. An audio signal of **10sin1000 πt** is used for AM with a carrier of

50 sin200000πt. Calculate:

i) Modulation index

ii) Required Bandwidth

iii) Total power

iv) Efficiency of AM [8] 2018 [F]

28. Differentiate encoding and modulation. Explain Differential PSK with an example. [8] 2018 [F]

29. An audio signal of 3 KHz is used for modulation with a carrier of 25 sin(2π×1000000t). Calculate:

i) Modulation index

ii) Required bandwidth

iii) Total power

iv) Efficiency of AM [7] 2018 [F]

30.Differentiate encoding and modulation. Explain Differential PSK with an example. [8] 2018 [S],2023[S]

31.An audio signal of 5sin500πt is used for AM with a carrier 25sin100000πt. Calculate:

i) Modulation index

ii) Required bandwidth

iii) Total power

iv) Efficiency of AM

32. Differentiate encoding and modulation. The sequence of the bit is 0101000011. What will be the output if the following encoding schemes are used?

i) AMI

ii) NRZ-L

iii) Bipolar 3 zero substitution (B3ZS)

iv) Differential Manchester [7] 2019 [F]

33. Encode the bit stream 10110001110 using the following scheme:

i) RZ

ii) NRZ-I

iii) NRZ-L

iv) AMI

v) Manchester [7] 2019 [S]

34.An audio signal of **$20\sin(2\pi\times800t)$** is used for AM with a carrier **$60\sin(2\pi\times10^6t)$.** Calculate:

i) Modulation index (Modulating)

ii) Required bandwidth

iii) Total power using load resistance

iv) Efficiency of AM [7] 2019 [S]

35. What is line coding? Represent the given sequence of bit 111000010 using:

i) AMI

ii) HDB3

iii) NRZ [7] 2020 [F]

36. Explain PCM as an A to D conversion technique with diagram. [5] 2020 [F]

37. Encode 1100110 using polar NRZ, Unipolar NRZ, Manchester and AMI technique. Also encode 10000000000110 using HDB3 and B8ZS scrambling technique. [10] 2020 [S]

38. What is line coding? Represent the given sequence of bits 1111000011 using i) NRZ-I, ii) Differential Manchester and iii) HDB3. [7] 2021 [F]

39. How an analog signal is converted into digital? Explain QPSK as a multilevel modulation with necessary diagram. [8] 2021 [F]

40. Differentiate between AM and FM. Why is FM superior over AM in communication? [8] 2021 [S],2024[F]

41. Draw ASK, FSK and PSK of the given binary sequence: 0100110011001010. [7] 2021 [S]

42. What is modulation? Explain QPSK modulation in detail. [7] 2022 [F]

43. How modulation helps to reduce the size of antenna? Illustrate in brief. Explain about amplitude, frequency, and phase modulation. [8] 2023 [S]

44. Differentiate between PSK, ASK and FSK modulation technique with necessary diagram. [8] 2023 [F] (old),2023 [F] (New)

45. Encode the bit stream of 011010001 using the following scheme:

i) NRZ-I

ii) NRZ-L

iii) Manchester

iv) Differential Manchester [7] 2023 (New) [F]

46.Explain Frequency modulation and phase modulation with waveforms of message signal, carrier signal and modulated wave. [8] 2024 [S]

47. What is multilevel modulation? Briefly explain with a suitable example. [8] 2024 [S]

48. How does delta modulation overcome the demerits of PCM? Briefly explain the working of delta modulation. [8] 2024 [S]

49. Briefly explain the digital modulation technique with suitable diagram. [8] 2024 [S]

50. What is a modem? Show Manchester encoding patterns for the bit stream 10111001101. [7] 2024 [S]

51. Represent the given sequence of bits 10110100 using [7] 2024 [F]

i) Unipolar Nonreturn –to-zero (NRZ)

ii) Polar NRZ

iii) Bipolar encoding

iv) Manchester encoding

52.Why do we need modulation in data communication? Explain. [7] 2024 [F]

53.Calculate the rate of a 500 baud 8-QAM signal with a constellation diagram. [7] 2024 [F]

54. Represent the given bit stream 10011100001101 using [8] 2024 [F] (New)

i) Polar NRZ

ii) Differential Manchester

iii) HDB3

55.Encode 01001101010110 using Manchester, AMI, NRZ, NRZ-I and explain line coding techniques. [8] 2025 [S]

56.What is multi-level modulation What is the importance of MODEM in digital communication system? [7] 2025 [S]

57.Explain FDM. State sampling theorem and why we need modulation in communication system? [7] 2025 [S] (New)

58. Define line coding. Explain the block diagram of PCM system. [7]2025 [S] (New)

59. Encode digital data 11010101011101 using [8] 2025 [S] (New)

i. Unipolar RZ
ii. Bipolar NRZ
iii. AMI
iv. Manchester